



Effects of Dust Deposition on the Surface Temperature and Performance of Photovoltaic Modules

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Abstract: Photovoltaic technology has become one of the most promising approaches for sustainable electricity generation; however, its performance is strongly influenced by environmental conditions, including dust accumulation and operating temperature. In this study, the combined effects of dust deposition and panel tilt angle on the thermal behavior and electrical performance of photovoltaic modules were experimentally investigated under controlled artificial illumination generated by light bulbs. Five irradiance levels (100, 200, 300, 400, and 500 W/m²) were employed to simulate different operating conditions, while various dust loading levels and panel tilt angles were systematically evaluated. The results demonstrated that the tilt angle significantly affected photovoltaic performance. As the dust loading increased, the panel's temperature rose. Although electrical power generation was successfully achieved through the bulbs, the output remained substantially low. Dust deposition reduced the amount of incident solar radiation reaching the photovoltaic modules, thereby decreasing electrical power generation and energy conversion efficiency. These findings provide valuable experimental evidence for optimizing photovoltaic system installation and maintenance in dusty environments and contribute to the development of more efficient and sustainable photovoltaic energy systems.

Keywords: Photovoltaic panels; Electrical energy; Light bulbs; Tilt angle; Efficiency

1 Introduction

As energy demands grow, renewable energy sources are gaining popularity as a solution [1]. To promote the use of renewable energy, governments worldwide are calling for policy changes [2–4]. As a result, the global installed renewable energy capacity is more than double that of 2011 by 2022 [5]. The most practical renewable energy source is solar power. It is also easily available, straightforward to install, and reasonably priced [6, 7]. Solar radiation transmits around 1.8×10^{11} MW of energy to the Earth's surface per second [8]. This number only makes up a small portion of the energy used worldwide [9]. Adinoyi and Said [10] found that photovoltaic modules tilted at 26° in eastern Saudi Arabia showed a 50% decrease in power output after six months of soiling. Furthermore, Benghanem et al. [11] found that modules tilted at 25° in the non-coastal region of western Saudi Arabia showed a 28% decline over two months. Photovoltaic module efficiency decreased by 37.6%, 14.1%, and 10.9% for 0°, 25°, and 45°, respectively, after 14 days of exposure to outdoor sunlight in Kuwait [12–17].

Mustafa et al. [18] investigated the effectiveness of the light spectrum from 10, 15, and 20 W light-emitting diode bulbs as a power source for producing electricity in a small monocrystalline solar module. Before the light spectrum was transported to the photovoltaic cell's surface, it was compared with and without a Fresnel lens. According to test results, photovoltaic cells exhibited different conversion efficiencies under the two configurations: an efficiency of 3.77% was obtained with the Fresnel lens at a 15 W lamp power, while an efficiency of 4.86% was achieved without the Fresnel lens at a 20 W lamp power.

Mustafa et al. [19] conducted a study to determine the photovoltaic tilt angle for harvesting photon energy from light-emitting diode lamps commonly used in homes or residential buildings. The tested solar panel tilt angles were 0°, 15°, 30°, and 90°, while the lamp position was fixed. The results showed that a 0° tilt produced the highest electrical power from the solar panels. The power output decreased to 16.33, 12.92, and 12.91 W at 15°, 30°, and

90°, respectively. The aim of studying the effect of tilt angle on the performance of solar panels is to understand the relationship between tilt angle and the efficiency of solar panels in converting solar energy into electrical energy. The purpose of the study is to use light bulbs to generate electricity and thus take advantage of the light energy emitted by the bulbs to convert it into electrical energy.

2 Materials and Methods

The experimental setup consisted of a 100-W polycrystalline photovoltaic module, a light-emitting diode lamp array, and the associated measurement instruments. The photovoltaic module was positioned in front of the light-emitting diode lamp array, which provided five irradiance levels of 100, 200, 300, 400, and 500 W/m². Figure 1 shows the entire experimental setup with all its components. The type of lamp used was a light-emitting diode, and the distance between the lamp array and the panels was 50 cm. Table 1 provides an overview of the technical specifications of the solar photovoltaic module and the measurement equipment used in this study.

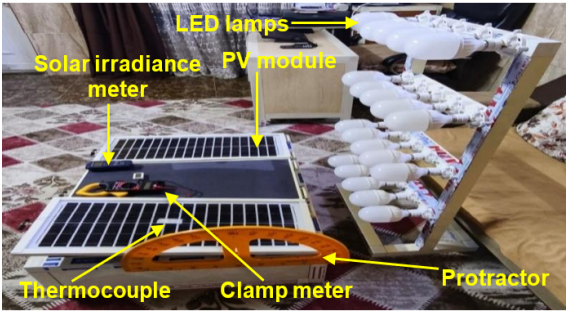


Figure 1. Entire experimental setup

Table 1. Technical specifications of the panel

Parameter	Value
Maximum Power Voltage (V_{mp})	18.2 V
Maximum Power Current (I_{mp})	5.49 A
Open Circuit Voltage (V_{oc})	22.1 V
Short Circuit Current (I_{sc})	5.93 A

3 Dust Used in This Study

Solar panels need to be cleaned regularly to ensure they operate at maximum efficiency. Dirt, dust, and other debris that accumulate on the surface of solar panels can reduce their ability to absorb sunlight and generate electricity. Therefore, dust is one of the most important factors affecting the performance of photovoltaic panels. There are many types of dust, such as soil, fertilizer, sand, and others. However, in this study, soil was used as the dust material because it is the most common type of dust found in nature that falls or accumulates on the panels and affects their performance. Four dust deposition densities of soil were used in this study to demonstrate the effect of the thickness and weight of the accumulated soil on performance (25, 50, 75, and 100 g/m²). Figure 2 shows the dust sample used in this study. For dust distribution, a very fine sieve was used to distribute the dust uniformly. Sieves (or “standard sieves”) are a device consisting of a mesh with specific-sized openings used to sort and separate particles according to their size. Sieves allow smaller particles to pass through while retaining larger particles (Figure 3). They are widely used, with opening sizes ranging from several microns to several millimeters. A sieve allows smaller particles to pass through while retaining larger particles and is determined according to specific standards such as ASTM or ISO.



Figure 2. Dust sample used in the study



Figure 3. Vibrating screen for dust distribution

4 Uncertainty Analysis

This section discusses the measurement uncertainties of the instruments (Table 2).

Table 2. Measurement uncertainty of the instruments

Instrument	Error (%)
Clamp meter	± 2
Pyranometer	± 5
Protractor	± 0.01
Thermocouple	± 0.2

5 Results and Discussion

5.1 Effect of Dust Deposition Density on the Surface Temperature of Photovoltaic Panels

Figure 4 illustrates the experimental effects of different dust deposition densities (25, 50, 75, and 100 g/m²) and tilt angles (10°, 20°, 30°, and 45°) on the surface temperature of the photovoltaic panel, compared with a clean panel. As the dust deposition density increased, the panel's surface temperature rose. This is because photovoltaic panels absorb sunlight and convert it into electricity, but only a portion of this light is converted into heat. As dust accumulated, heat absorption increased, thereby increasing the panel surface temperature. The study showed that the temperature at 30° is 10 °C higher than at 20° under 100 g/m².

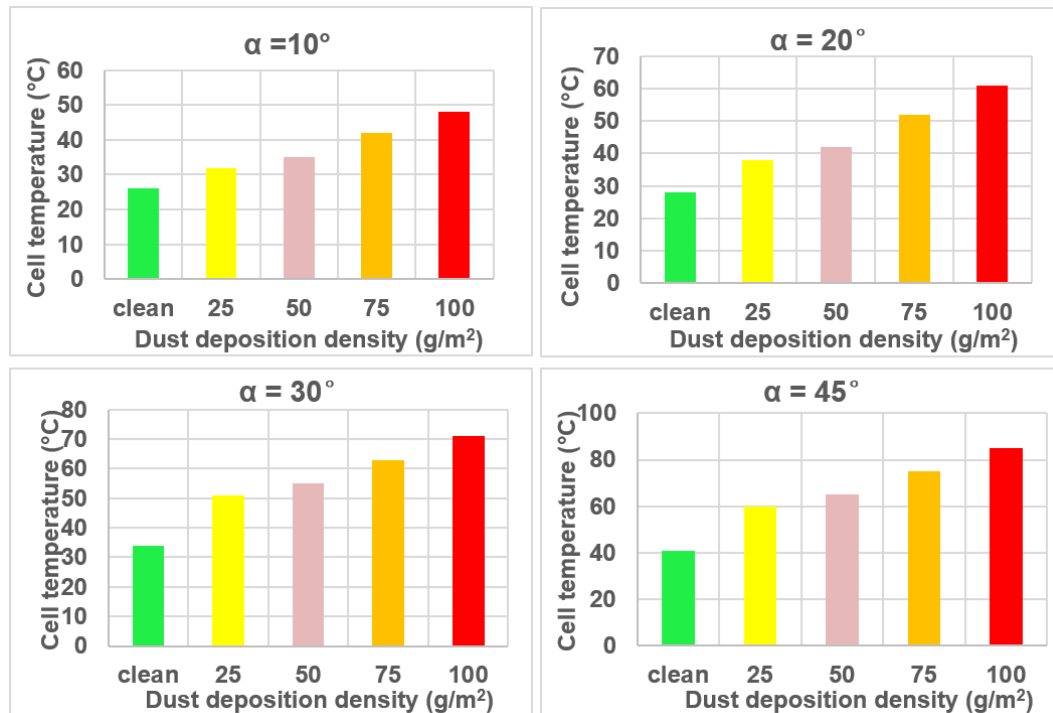


Figure 4. Effect of dust deposition density on the surface temperature of the photovoltaic module at different tilt angles under an irradiance of 500 W/m²

5.2 Effect of Dust Deposition Density on the Power Output of Photovoltaic Panels

Figure 5 illustrates the experimental results of the effect of different dust deposition densities accumulating on the panel surface (25, 50, 75, 100 g/m²) at different angles (10°, 20°, 30°, 45°) on the maximum power output. As the dust deposition density increased, the panel's power output decreased. This is due to the reduced amount of light reaching the photovoltaic cells, leading to lower electrical output. The results showed that power increased as the panel tilt angle increased by 10% to 40% due to the resulting perpendicularity between the incident light rays and the solar panel, thus reducing radiation wasted due to angular deviation. The results also indicated that increasing the dust deposition density led to a power decrease of 30% to 50%.

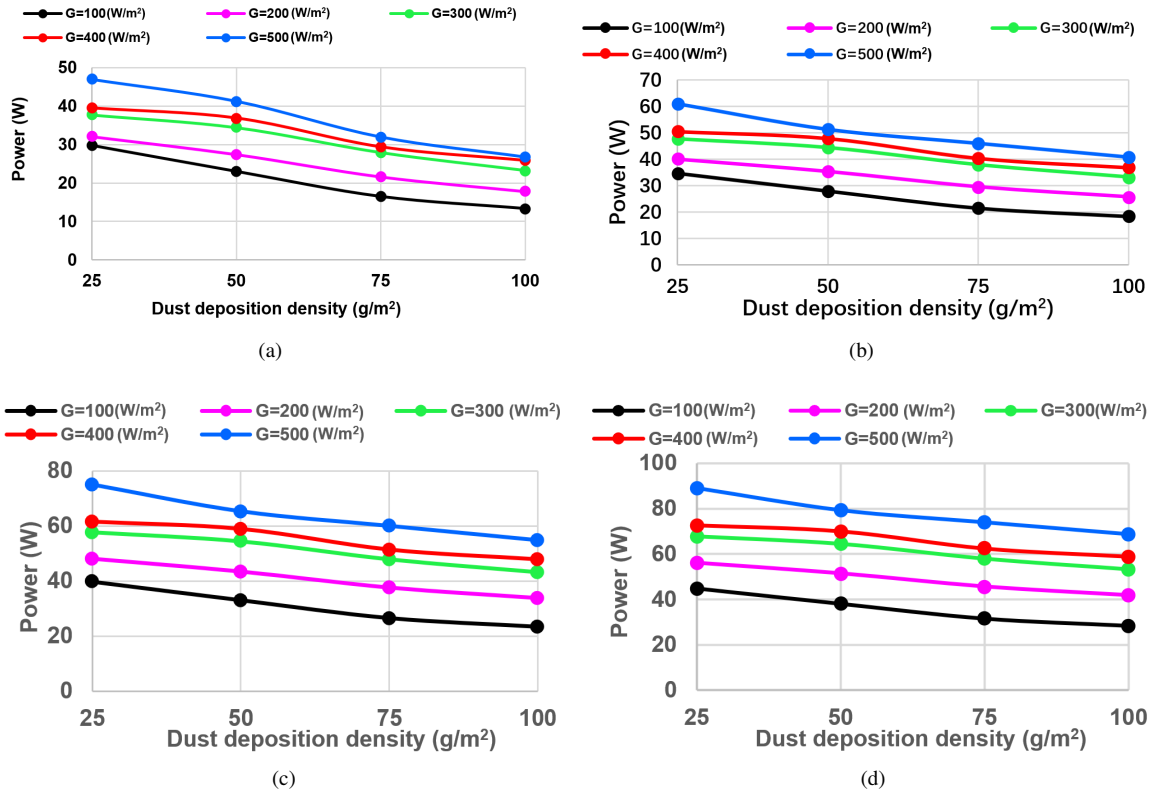


Figure 5. Effect of dust deposition density on the maximum power output of the photovoltaic module under different irradiance levels: (a) $\alpha = 10^\circ$; (b) $\alpha = 20^\circ$; (c) $\alpha = 30^\circ$; (d) $\alpha = 45^\circ$

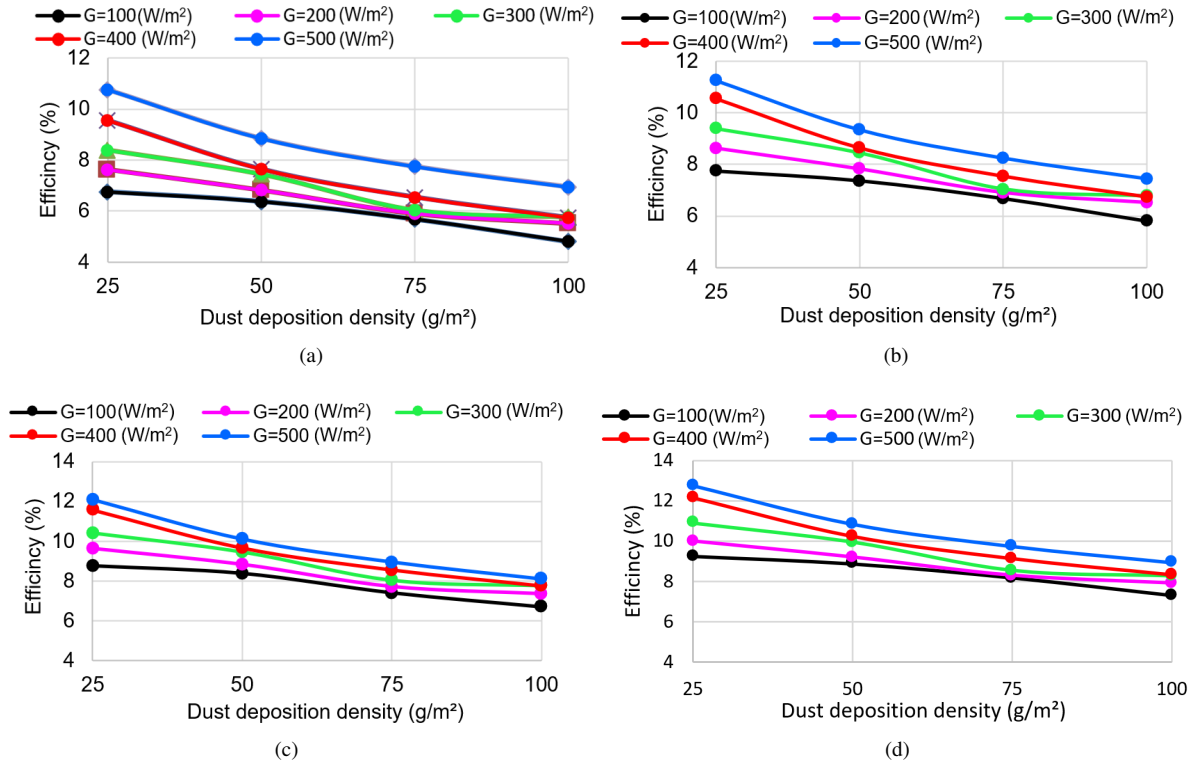


Figure 6. Effect of dust deposition density on the conversion efficiency of the photovoltaic module under different irradiance levels: (a) $\alpha = 10^\circ$; (b) $\alpha = 20^\circ$; (c) $\alpha = 30^\circ$; (d) $\alpha = 45^\circ$

5.3 Effect of Dust Deposition Density on the Efficiency of Photovoltaic Panels

The dust deposition density on photovoltaic panels was inversely proportional to the energy output. As the accumulated dust (or dust deposition density) increased, the panel's electrical efficiency and power generation decreased linearly, leading to total energy losses that ranged from 20% to more than 50%. The increased dust deposition density on photovoltaic panels significantly affected the efficiency of the panels in converting solar energy into electricity. Figure 6 illustrates the effect of different dust deposition densities (25, 50, 75, 100 g/m²) and different angles (10°, 20°, 30°, 45°) on the conversion efficiency of the photovoltaic module. The greater the dust deposition density, the lower the conversion efficiency. This reduction occurs because dust blocks part of the incident light. The thicker the dust on the panels, the more light is blocked, thus reducing the amount of energy generated, which in turn reduces the conversion efficiency due to the direct relationship between power and efficiency. At lower densities (25 g/m²), an efficiency drop of around 33% was observed, and at heavier accumulations ranging from 50 g/m² to 100 g/m², power losses exceeded 40% and reached approximately 50%, depending on the type of dust and local climate.

6 Comparison with Previous Research

Table 3 presents some studies published in the scientific literature and compares them with the present study. Previous studies have extensively investigated the effects of dust deposition on the performance of photovoltaic modules. However, the literature review indicated several factors affecting the performance of the photovoltaic system and power generation, such as location, dust characteristics, and pollution in the area.

Table 3. Summary of some published studies in the literature

Reference	Year	Country	Power Loss (%)
[13]	2013	Belgium	3–4
[14]	2013	India	0.33
[15]	2015	Qatar	10–20
[16]	2015	Poland	6.24
[17]	2016	Iraq	18.74
[18]	2017	Pakistan	20
[19]	2018	Iran	21.47
[20]	2018	China	34
This study	2026	Iraq	30–50

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7 Comparative Analysis and Conclusion

This study investigated the effects of panel tilt angle, dust deposition density, and operating temperature on the electrical performance of photovoltaic modules using light bulbs. The influences of these factors on the current, voltage, power, and efficiency were experimentally evaluated. The main findings are summarized as follows:

- It is possible to generate energy by utilizing the light of the lamps as solar radiation.
- Since voltage tends to decrease with increasing temperature and current may also be affected, the output power generally decreases with increasing temperature.
- Sustainable development and environmental conservation can be supported through energy generation from alternative sources such as lamps.
- At lower densities (25 g/m²), an efficiency drop of around 33% is expected, and at heavier accumulations ranging from 50 g/m² to 100 g/m², power losses exceed 40% and reach approximately 50%, depending on the type of dust and local climate.
- The results also indicate that increasing the dust deposition density can lead to a power decrease of 30% to 50%.
- The study shows that at a dust deposition density of 100 g/m², the panel temperature at 30° was 10 °C higher than that at 20°.

Author Contributions

Conceptualization, E.H.; methodology, E.H.; investigation, E.H.; formal analysis, E.H.; writing—original draft preparation, S.R.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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